The dimension-nine threshold: an exact reduction and minimality for constructions from paired seeds

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Abstract

The accompanying paper proves separation from the point-generated product–join class in every dimension at least nine. We show that global minimality of nine is equivalent to the unrestricted simplex bound in dimension eight, which remains unproved here. We do prove that nine is the first separation dimension within the paired two-upper-facet seed family and for all constructions from those seeds by products and joins. The proof uses a rational supporting gradient and log-concavity; a separate positive-polynomial certificate checks the seed bound. Neither argument reduces arbitrary convex bodies to these constructions.

1 What global minimality requires

For a full-dimensional convex body $K \subset \mathbb{R}^n$, put

$$R_n(K) = \frac{\mathrm{vol}_n(\Pi K)}{\mathrm{vol}_n(K)^{n-1}}, \qquad g(n) = \frac{n^n}{n!}, \qquad c_n = (n+1)g(n).$$

Here ΠK is the projection body. Let $\mathcal{C}$ be the class generated from a point by Cartesian products, joins, and invertible affine maps on the relevant affine hulls. The exact finite classification in [3] gives $\sup_{K \in \mathcal{C}_n} R_n(K) = c_n$ for $1 \le n \le 9$. The accompanying paper [2] supplies a strict witness in dimension nine and separation in every larger dimension. The unrestricted simplex bound in dimensions four through eight is left open in [1, Introduction].

Proposition 1. *Nine is the smallest dimension admitting separation from $\mathcal{C}$ if and only if*

$$R_8(K) \le c_8 = \frac{131072}{35} \quad \textit{for every full-dimensional convex body } K \subset \mathbb{R}^8. \tag{1}$$

Proof. The inherited pyramid calculus [4, Sections 2–5] gives, for a d-dimensional convex body K and $t \ge 0$,

$$\frac{R_{d+t}(\mathcal{P}^t K)}{c_{d+t}} = \frac{R_d(K)}{g(d)} \frac{1 + t a(K)}{d+t+1}, \qquad a(K) \ge \frac{1}{d+1}.$$

Here $a(K)$ is the lifted facet invariant of [4], extended by continuity to convex bodies. The difference between consecutive terms on the right is

$$\frac{R_d(K)}{g(d)} \frac{(d+1)a(K) - 1}{(d+t+1)(d+t+2)} \ge 0.$$

Thus any strict violation of the simplex bound in a dimension $d \le 8$ lifts to a strict violation in dimension eight. If (1) holds, no such lower-dimensional separation exists, and the known dimension-nine witness proves minimality. Conversely, a violation of (1) is already separation in dimension eight. □

By continuity of volume and the projection-body operator under Hausdorff convergence, it would suffice to prove (1) for every full-dimensional rational polytope in $\mathbb{R}^8$. This is still an unbounded class; finite searches do not establish that assertion.

2 A proved minimum for the paired family

For $0 \le u_1 \le \cdots \le u_m < 1$ and $n \ge 2m$, define

$$K_n(u) = \{x \in \mathbb{R}^n : x_i \ge 0,\ \mathbf{1}\cdot x + |b \cdot x| \le 1\}, \qquad b = (-u_m, \dots, -u_1, u_1, \dots, u_m, 0^{n-2m}).$$

For $m = 0$ this means the standard simplex. The family is inherited from the two-facet constructions of [1]; we make no priority claim for the family or the dimension-nine phenomenon.

Theorem 2. *Every paired seed $K_n(u)$ in dimension $n \le 8$ satisfies $R_n(K_n(u)) \le c_n$. The paired seed $K_9(1/100, 2/100, 3/100, 5/100)$ satisfies*

$$\frac{R_9}{c_9} = \frac{20811283583}{20797677600} > 1.$$

Consequently nine is the first separation dimension within this paired seed family, with zero coordinates and affine images allowed.

We first record an elementary consequence of log-concavity.

Lemma 3. *If $\mathbb{P}(Y > 0) = 1$, Y has a log-concave survival function, and its mean M is finite, then $\mathbb{E}e^{-Y} \le (1 + M)^{-1}$.*

Proof. Write $S(t) = \mathbb{P}(Y > t)$, so $S(0) = 1$. Convexity of $-\log S$ implies $S(t + s) \le S(t)S(s)$ for $s, t \ge 0$. Consequently

$$J(t) := \int_t^\infty S(s)\, ds \le MS(t).$$

For $L = \int_0^\infty e^{-t}S(t)\, dt$, Tonelli's theorem gives $\int_0^\infty e^{-t}J(t)\, dt = M - L \le ML$. Hence $L \ge M/(1 + M)$ and $\mathbb{E}e^{-Y} = 1 - L \le (1 + M)^{-1}$. □

Proof of Theorem 2. It is enough to consider dimension eight: appending a zero tilt is a pyramid, so Proposition 1's monotonicity transfers the eight-dimensional bound to smaller dimensions. Padding with zero pairs then gives four parameters $0 \le u_1 \le \cdots \le u_4 < 1$. Let X_i be independent standard Laplace variables, each with density $e^{-|x|}/2$, and write

$$M(u) = \mathbb{E}\left|\sum_{i=1}^4 u_i X_i\right|, \qquad I(u) = 8!\,\mathrm{vol}_8(K_8(u)).$$

The homogeneous gauge integral gives

$$I(u) = \int_{\mathbb{R}^8_+} e^{-\mathbf{1}\cdot x - |b\cdot x|}\, dx = \mathbb{E}e^{-Y}, \qquad Y = \left|\sum_{i=1}^4 u_i X_i\right|,$$

because the difference of two independent standard exponential variables is standard Laplace. If $u \ne 0$, the sum has a symmetric log-concave density by Prékopa's convolution theorem [5, Theorem 7]. The density of Y on $[0, \infty)$ and its survival function are therefore log-concave; the latter follows by integrating $f(t + s)$ over $s \ge 0$ using [5, Theorem 6]. Lemma 3 yields $I(u) \le (1 + M(u))^{-1}$.

The convex homogeneous function M on $\mathbb{R}^4$ has, at $v = (1, 2, 4, 8)$, supporting gradient

$$\nabla M(v) = \frac{1}{18225}(2749, 5304, 9624, 15664). \tag{2}$$

For a direct exact calculation, partial fractions of the characteristic function $\prod_i (1 + u_i^2 t^2)^{-1}$ give

$$M(u) = \sum_{i=1}^4 \frac{u_i^7}{\prod_{j\ne i}(u_i^2 - u_j^2)} \quad (0 < u_1 < \cdots < u_4).$$

Differentiating at v gives (2). Convexity and homogeneity imply $M(u) \geq \nabla M(v) \cdot u$. Writing $x_1 = u_1$ and $x_j = u_j - u_{j-1}$ for $j \geq 2$, we obtain

$$M(u) - \frac{u_1 + 3u_2 + 5u_3 + 7u_4}{9} \geq \frac{941x_1 + 217x_2 + 988x_3 + 1489x_4}{18225} \geq 0. \tag{3}$$

The seed formula in [2, Proposition 2] now gives, with $w \cdot u = u_1 + 3u_2 + 5u_3 + 7u_4$,

$$\frac{R_8(K_8(u))}{c_8} = I(u)\left(1 + \frac{w \cdot u}{9}\right) \leq \frac{1 + w \cdot u/9}{1 + M(u)} \leq 1.$$

For $u = 0$ the body is a simplex; repeated and zero parameters also follow by continuity, or directly from the integral argument. The displayed rational value in dimension nine is checked by the exact seed formula and the independent geometric certificate in [2]. Since the class maximum there is c_9, this is separation from $\mathcal{C}_9$. □

2.1 Closure under arbitrary products and joins

Let $\mathcal{F}$ be the class generated by a point and all paired seeds $K_n(u)$, using Cartesian products, joins, and invertible affine maps. This allows arbitrary finite operation trees, rather than only pyramids or the balanced recursion of the accompanying paper.

Proposition 4. *For $1 \leq n \leq 8$ and every $K \in \mathcal{F}_n$, one has $R_n(K) \leq c_n$. Thus nine is the first separation dimension within $\mathcal{F}$.*

Proof. Use the inherited coordinates $H = 1/a$ and $Q = aR/g(n)$ from [4]. The universal invariant bounds give $1 \leq H \leq n + 1$. We prove by induction on operation trees of output dimension at most eight that

$$Q(H + 8 - n) \leq 9. \tag{4}$$

The formal point state is $n = 0$, $H = Q = 1$, which satisfies the bound. For a seed of dimension $n \leq 8$, the left side divided by nine is $R_8(\mathcal{P}^{8-n}K)/c_8$. These pyramids remain paired seeds, so Theorem 2 proves the base cases.

For a join of factors with dimensions r, s, states $(h, q), (j, v)$, and $n = r + s + 1 \leq 8$, the new state is $(h + j, qv)$. Put $A = h - r - 1 \leq 0$ and $B = j - s - 1 \leq 0$. Then

$$(h + 8 - r)(j + 8 - s) = (9 + A)(9 + B) \geq 9(9 + A + B) = 9(h + j + 8 - n).$$

The two input bounds therefore imply the output bound.

For a product, assume $1 \leq r \leq s$ and $n = r + s \leq 8$; products with a point do not change a state. With $G = g(r)g(s)/g(n)$, the operation rules give

$$H = \frac{nhj}{rj + sh}, \qquad Q = qvG\frac{rj + sh}{n}.$$

It suffices to verify

$$(h + 8 - r)(j + 8 - s) - 9G\left(hj + \frac{8 - n}{n}(sh + rj)\right) \geq 0. \tag{5}$$

This is bilinear in (h, j), so its minimum on $[1, r + 1] \times [1, s + 1]$ occurs at a corner. For the sixteen pairs $1 \leq r \leq s$, $r + s \leq 8$, all 64 corner values are at least nine; the checker recomputes each exact value. This proves product closure.

Finally, for $n \leq 8$, inequality (4) implies

$$\frac{R_n(K)}{c_n} = \frac{HQ}{n + 1} \leq \frac{9H}{(n + 1)(H + 8 - n)} \leq 1,$$

where the last inequality is $(8 - n)H \leq (8 - n)(n + 1)$. The known paired witness in dimension nine belongs to $\mathcal{F}$ and exceeds the class maximum c_9. □

3 Independent arithmetic check and remaining scope

The file `check_minimality.py` recomputes the supporting gradient, the four margins in (3), the dimension-nine rational ratio, and every corner obligation in (5). An unpruned enumeration of all operation states gives the simplex maximum in dimensions one through nine, independently cross-checking the previously published Pareto classification [3].

There is also a separate algebraic proof of the eight-dimensional family bound. For distinct positive parameters, write

$$I(u) = \frac{N(u)}{D(u)}, \qquad D(u) = \prod_i (1 + u_i) \prod_{i<j} (u_i + u_j).$$

The supplied integer numerator N has 125 monomials. The checker reconstructs D and verifies the divided-difference formula for I after multiplication by the Vandermonde polynomial. It then substitutes $u_i = x_1 + \cdots + x_i$ into

$$9D(u) - (9 + w \cdot u)N(u).$$

The resulting polynomial has 585 nonzero monomials, all with positive integer coefficients, of total degrees seven through ten. This proves the bound throughout the ordered positive parameter region; continuity handles its boundary. The checker uses standard-library integer and rational arithmetic, with no computer-algebra system or optimizer.

Run the following commands from this supplement's directory:

```
python -B check_minimality.py --check-report
python -B -m unittest -v test_minimality.py
python -B -O -m unittest -v test_minimality.py
```

The tests reject altered numerators and denominators, false or missing gap coefficients, duplicate monomials, and invalid exponents. The analytic arguments are supplied above, rather than inferred from finite checks, and have not been proof-assistant formalized or externally peer reviewed.

Theorem 2 and Proposition 4 exclude the paired family and its complete product–join closure as sources of a lower-dimensional witness. General asymmetric seeds and arbitrary convex bodies need not belong to this generated class. The results do not prove (1), nor imply that every maximizer belongs to $\mathcal{F}$. Global minimality is therefore open in this work. The next substantive requirement is a bound covering all eight-dimensional convex bodies, or an exact counterexample refuting it.

Generative AI disclosure

OpenAI Codex agents contributed substantially to source discovery, literature synthesis, and mathematical exposition.

Data and code availability

This note, its source, polynomial certificate, checker, and exact-arithmetic report are supplied in the accompanying reproduction repository [2], under `research/minimality/`. The original paper and its versioned release remain separate from this research supplement. The supplied checks establish the stated arithmetic obligations; they do not claim an unrestricted nonexistence proof or global bibliographic priority.

References

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